B1

1. Describe the modern periodic law ?

Sol^n: The physical and chemical properties of the elements can be changed in accordance with their atomic number / electronic configuration

Characteristics of the modern periodic table:

01. The elements are arranged in a sequence in the modern periodic table.

02. There are 7 horizontal & 18 vertical coloumn in the modern periodic table

03. The horizontal lines are called the period and the vertical coloums are called the group.

2. Write the classification of elements on the basis of the electronic structure ?

Sol^n: According to the outer electronic configuration there are 4 kinds of elements.

Such as —
ⓐ s — block
ⓑ p — block
ⓒ d — block
ⓓ f — block

s-block element : If the electrons enter into outermost energy level of atom to s orbital in accordance with their electron capacity is called the s — block elements

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For example:

$Na(11)$ —— $1s^2 2s^2 2p^6 3s^1$

$Ca(20)$ —— $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$

These elements are known as base metal.

02. p-block element: If the electrons enter into the outermost energy level of atom into p orbital in accordance with their electron capacity is called the p— block element.

for example:

Al_{13} —— $1s^2 2s^2 2p^6 3s^2 3p^1$

Cl_{17} —— - - - - —— $3p^5$

Kr_{36} - - - - - - $4s^2 4p^6$

03, d- block element: If the electrons enter into the outermost energy level of atom into d— orbital in accordance with their electron capacity is called the d— block element

In generally the elements of B— sub group are called the d— block element

The electronic configuration of these elements is: $(n-1)d^{1-10} ns^{1-2}$

All the elements are metal

the electronic configuration of transition element is $(n-1)d^{1-9} ns^{1-2}$

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04. If the electrons enter into the outermost energy level of atom into ~~of~~ orbital in accordance with their electron capacity is called the f-block element In generally the f orbital is in complete for electron $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$

Q, What is "transition element" ? give their properties in accordance with the periodic table.

Soln: Elements which have normally the same number of electrons in the outermost level but have a progressively greater Number of electrons in an inner level (such as d level) are called "Transition Element". or the elements of sub group-B and VIII is called the transition elements.

Properties of transition elements: 01. All the elements are high melting points, electropositive and heavy metals.

02. These metals have almost the same atomic and ionic sizes

03. All these elements show positive oxidation states of +2 and +3 generally and form mostly ionic compounds.

04. As a general rule, the transition element form coloured compounds

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05. These elements are also effective catalytic agents.

06. All these form quite a large number of complex compounds.

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Common Definition:

(i) Representative Elements: The elements of sub group A is called the representative elements. The electronic configuration of these element is $ns^{1-2}np^{0-5}$

(ii) Transition Elements: The elements of sub-group III and VIII is called the transition-element, incomplete d-orbital is the main characteristic of this element. The electronic configuration of this element is $(n-1)d^{1-9}ns^{1-2}$

(iii) Inner-Transition Element: The elements which have 4f and 5f orbital in electronic configuration is called the Inner-transition element. The electronic configuration of these elements is $(n-2)f^{1-14}(n-1)d^{0-2}ns^{2}$

4f elements are called the Lanthanide elements and 5f elements are called the Actinide elements

(iv) Inert gasses: The elements of zero group is called the Inert gases. The electronic configuration of this gases is ns^2np^6.

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Alkali Metals : Except hydrogen the electronic configuration of IA group is ns^1 riched metals are called the Alkali-metal.

(VI) Alkaline-Earth Metals : The electronic configuration ns^2 of IIA group elements are called the Alkaline Earth-Metals.

(VII) Coinage Metals : The electronic configuration $(n-1)d^{10}ns^1$ of IB group elements are called the coinage Metals. Cu, Ag, Au

(IX) Rare Earth Metals: The electronic configuration 4f electron riched metals are called the rare earth elements.

(X) Halogens : The electronic configuration ns^2np^5 of VIIA group elements are called the Halogens elements.

Q. Describe the position of Hydrogen in the periodic Table?

Soln : The position of Hydrogen in the periodic table is left undecided. It has similarities in properties with both the alkali metals and the halogens.

(A) Position of H with Alkali metal of IA :

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① Electronic configuration: H has 1 electron in S orbital like the alkali metals of 1A

$H\ (1) \longrightarrow 1s^1$

$Li\ (3) \longrightarrow 1s^2 2s^1$

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(ii) Oxidation Number and electronegativity: For ns1 electronic configuration of alkali metal 1A group H represent oxidation Number +1

$Na - \bar{e} \longrightarrow Na^+$

$H - \bar{e} \longrightarrow H^+$

(iii) Mono covalent: Like Lithium, Hydrogen creates covalent bond and its mono covalent is one. For example. CH_4, $LiCH_3$, LiC_2H_5, C_2H_6

(iv) Hallaide Formation: Like alkali metals H creates hallaide compound. For example.

HF, HCl, HBr, HI

($\equiv$ NaF, NaCl, NaBr, NaI)

(v) Reduction properties: Like alkali metals H also a strong oxidant.

(B) Position of H with VII A group:

(1) electronic Configuration: H has 1 less electron near the inert gass. Hence the hallogens has 1 electron near the inert gass.

H_1 : 1 electron

He_2 : 2 electron

For example

Cl_{17} : 2, 8, 7 electron

Ar_{18} : 2, 8, 8 electron

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ion metal properties: like hallogens H is a non-metal.

(iii) Physical properties: H is a leght gass and F, Cl is a heavy gass.

(iv) Dieal-atomic properties: like hallogen H is also a dual atomic element.

(v) Vallency: Like hallogen H is also a mono-vallency element.

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Q. Describe the Usefulness of Periodic Table?

Soln: The physical and chemical properties of the elements are periodic function of their atomic number. The vertical arrangements of elements in periodic Table are called groups. And the horizontal arrangements of the elements in periodic Table are called period.

Usefulness of Periodic Table:

(01). Classification of the elements: The classificateon of elements of similar properties in to groups simplified their study.

(02) Prediction of undiscovered elements: with the help of periodic table it is possible to predict the undiscovered elements.

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03. Correction of atomic weight: With the help of periodic table it is possible to correct the atomic weight of the element.

04 Periodic Table for industrial Research:
The periodic table has been found to be quite useful in industrial researches.

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M2 Che periodic 5

8. What is Diagonal Relationship? Describe about it?

Soln: Li of group I A resembles Mg of group II A in many respects contrary to its group properties. Similar relationship exists between the three elements Be of group II A, Aluminium of group III A, Boron of group III A shows likeness with the silicon of group IV A. This is called the diagonal relationship.

top to bottom

	I A group	II A	III A	IV A
1st	H			
2nd	Li	Be	B	C
3rd	Na	Mg	Al	Si

Physical & Chemical

1 mark

Explanation: With the help of the electropositivity we can explain the diagonal relationship.

01. If we pass left to right in one period the electropositivity will decrease and increase in the group.

02. In the other hand, we can explain the diagonal relation with the help of ionic radius or power polarisation

In the same period from left to right the ionic radius will decrease but increase the positive charge on nucleus.

So, the polarization power will increase from left to right. In the same group the ionic radius will increase and decrease the polarization power.

(V)

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Q. What is Ionization Energy or potential? Describe the factors of it?

Soln: The ionization energy (IE) of an element is defined as the energy needed to remove a single electron from a atom of the element in the gaseous state. For example.

$$M(g) + IE \longrightarrow M^{+}(g) + \bar{e}$$

* The first ionisation energy (IE_1) is the energy needed to remove the first electron from the gaseous atom M to form M^+ ion
* The second ionisation energy, (IE_2), is the energy needed to remove a second electron from the gaseous M^+ ion to form M^{+2} ion.

$$M(g) + IE_1 \longrightarrow M(g) + \bar{e}$$
$$M^{+}(g) + IE_2 \longrightarrow M^{+2}(g) + \bar{e}$$
$$M^{+2}(g) + IE_3 \longrightarrow M^{+3}(g) + \bar{e}$$

1-30

Ca(20)
Mg

Factors of ionisation energy:

(i) Radius of the atom
(ii) Effective nuclear charge
(iii) Electronic configuration of atom
(iv) Screening effect
(v) Type of outermost electron
(vi) Number of positive charge or Catalon
(vii) Form the electronic configuration of inert gass.

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Q. What is electron affinity? Describe the common factors of it?

Soln: Electron affinity (EA) of an element is the amount of energy released when an electron is added to a gaseous atom to form anion.

$$X(g) + \bar{e} \longrightarrow X^-(g) + EA$$

The energy involved in the addition of the first electron is called first electron affinity. the energy involved in the addition of a second electron is called second electron affinity; and so on.

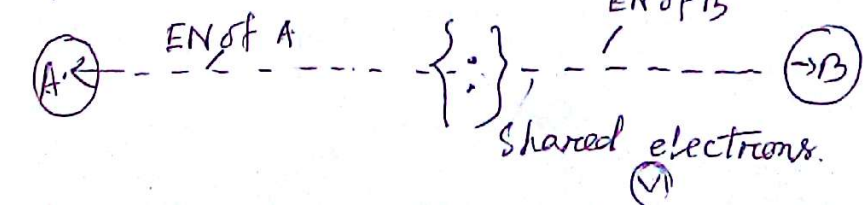
$$X + \bar{e} \longrightarrow X^- + EA_1$$
$$X^- + \bar{e} \longrightarrow X^{2-} + EA_2$$

Factors of electron affinity:

(i) Radius of the atom.
(ii) Effect Nuclear charge
(iii) electronic configuration of the atom
(iv) Number of Negative charge or anion.

Q. What is Electronegativity? Describe about it?

Soln: The attraction exerted by an atom on the electron pair bonding it to another atom by a covalent bond



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Importance of Electronegativity:

(I) In predicting the polarity of a particular bond

(II) In predicting the degree of ionic character of a covalent bond

(III) In Predicting of inductive effect in organic Chemistry.

(IV) In understanding the shape of molecules.

** In the same group from top to bottom in the periodic table the value of electron affinity decrease

** In the same period from left to right in the periodic table the electron affinity will increase.



H_2O

1.5 - 1.7

H - - - O - - - H -

NaCl

Na Cl

Why it is difficult to create K^{+2}? Explain with example?

Soln. The value of 1st (E_1) and 2nd (E_2) Ionisation of Potassium is given below.

$$K_{(g)} + E_1 \longrightarrow K^+_{(g)} + \bar{e} \; ; E_1 = 418 \text{ kJ mol}^{-1}$$

$$K^+_{(g)} + E_2 \longrightarrow K^{2+}_{(g)} + \bar{e} \; ; E_2 = 3070 \text{ kJ mol}^{-1}$$

It is impossible to gain the 2nd ionisation energy of Potassium (3070). By giving an electron K form the electronic configuration of Ar, which follow the octed rule. It is difficult to remove a second electron from it.

So, it is impossible to create the dual charge of K^{+2}.

Why the value of ionisation decrease in the same group from top to bottom? Explain ——

or,

[The atomic Number of K (19) is greater than Na (11) Hence the ionisation energy of K (418 kJ mol^{-1}) is less than Na (496 kJ mol^{-1})]

Soln. Electronic configuration of Na(11) and K(19) is given bellow——

$$Na_{(11)} \longrightarrow 1s^2 2s^2 2p^6 3s^1$$

$$K_{(19)} \longrightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 4s^1.$$

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In the same group from top to bottom in the periodic table, in every step increase the principal quantum number. So the radius is increase. for a long distance the attraction will decrease and it is easy to remove an electron. So the value of ionisation will decrease,

In the electronic configuration of Na (11) there are 3 shell and for K (19) there are 4 shell.

for Na (11) it is difficult to remove an electron than to K (19)

So, the ionisation value of Na is greater than the value of K (19).

Q. Explain with example, the ionisation value of Ca (20) is greater than K (19)?

Q. the ionisation value of Be (4) is greater than to B (5)? Explain ——

Q. Why the 1st electron affinity of O_2 is Negative and the 2nd electron affinity of O_2 is positive?

Solⁿ: $O_{(g)} + \bar{e} \longrightarrow \bar{O}_{(g)} + A_1$

$A_1 = -141$ kj mol^{-1} (exothermic)

$\bar{O}_{(g)} + \bar{e} \longrightarrow O_{(g)}^{-2}$ (Endothermic)

$+ A_2$

$A_2 = +780$ kj mol^{-1}

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10. Q. Write short Notes on diagonal relationship of Elements in periodic table.

Soln: The former sheet.

Q. Draw the position of H- in the periodic table?

Soln: See the former sheet.

11. Q. The first ionisation energy of Be is higher than that of B. Explain why?

Soln. The electronic configuration of Be and B is given below

B (5) ⟶ $1s^2 2s^2 2p^1$

Be (4) ⟶ $1s^2 2s^2$

The Ionization energy of Be is higher that that of B, Because the 2p¹ electron of Boron (B) is needed a small amound of energy to remove it in gaseous atom to a gaseous cation., but from the $2s^2$ electron of Be, one electron is needed a large amound of energy to remove it in gaseous atom to a gaseous cation. Remove one electron from gaseous atom to a gaseous cation is called 1st ionization potential. So we say the first ionization energy of Be is higher than that of B.

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12. The first ionization potential or energy of N is greater than that of oxygen. Explain why.

Soln: The electronic configuration of N and O is given below.

$O(8) \longrightarrow 1s^2 2s^2 2p_x^2 2p_y^1 2p_z^1$

$N(7) \longrightarrow 1s^2 2s^2 2p_x^1 2p_y^1 2p_z^1$

The ionization energy of N is greater than that of oxygen. Because from the $2p_x^2$ electron of oxygen is needed to small amount of energy to remove it in gaseous atom to a gaseous cation. But $2p_x^1$ electron in Nitrogen is needed to a large amount of energy to remove it in gaseous atom to a gaseous cation. So we say that the first ionization energy of N is greater than that of O.

13. ** Explain why the second ionization potential of an element is much higher than that of first one.

Exam: 2001, 2004.

Soln: The energy required to remove one more electron from the gaseous cation $M^+(g)$ to get the double positively charged gaseous cation M^{+2} is called the second ionization potential.

$$M^+(g) + I_2 \xrightarrow{-e} M^{+2}(g) + \bar{e}$$

The second Ionization potential is greater than

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that of the 1st one it can be explained effective nuclear charge.

We know that the cations are formed by removing the electrons from the atom / cation one by one.

$$M_{(g)} \xrightarrow{-\bar{e}} M^{+}_{(g)} \xrightarrow{-\bar{e}} M^{+2}_{(g)} \xrightarrow{-\bar{e}} M^{n+}_{(g)}$$

Thus a given cation has lesser number of electrons than its parent atom or cation. With the decrease of the number of electrons, the magnitude of the effective nuclear charge increases from $M^{+}_{(g)}$ to $M^{+2}_{(g)}$. With the increase of effective nuclear charge the magnitied of the force of attraction between the nucleus and the outer-most shell electron could also increases.

With the increase of force of attraction, the magitude of energy to remove the outer-most shell electron also goes on increase from $M^{+}_{(g)}$ to $M^{+2}_{(g)}$. Hence the second ionization energy is greater than the first one.

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*Q. What is electron affinity?

Ans. See the sheet.

14. Q. Explain why the 1st electron affinity of an element is usually an exothermic process while the 2nd one is endothermic process.

Example: 2000, 2005.

Soln. The amount of energy released in addition extra electron from outside to an isolated gaseous atom is called the 1st electron affinity.

Thus,

$M_{(g)} + \bar{e}_{(g)} \longrightarrow M^-_{(g)} + \text{Energy released}. \quad (= -E)$

Since in the addition of an electron energy is released. So released the energy is exothermic process.

Second electron affinity is defined as the amount of energy required to add one more electron to its mono negative anion $M^-_{(g)}$ to convert it into di-negative anion $M^{2-}_{(g)}$

Thus, $M^-_{(g)} + \bar{e} \longrightarrow$ Energy supplied $(= +E_2) \longrightarrow M^{2-}_{(g)}$

In this process of adding one more electron to $M^-_{(g)}$ anion against the electrostatic repultion between the extra electron. (Supplied energy is endothermic process) being added to $M^-_{(g)}$ and the negative charge on $M^-_{(g)}$ anion, energy instead of being released is supplied to $M^-_{(g)}$ ion to convert into $M^{2-}_{(g)}$. Thus the process

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of adding a second electron to $M^-(g)$ anion is an endothermic process.

15. Q. F has lower electron affinity than Cl. Explain

Exam: 2001, 03, 04

Soln: F has lower electron affinity than Cl. Because the addition of an extra electron to F atom, produce high electron density round the resulted anion. This high electron density increases, the repulsion between the electrons already present in the relatively compact 2p orbital of the 2nd shell of F atom and the electron being added. Due to this electron-electron repulsion, the F show lesser tendency to attract the extra electron from outside and hence F has lower electron affinity than Cl

16. Q. The electronegativity of halogen's decrease in the order F > Cl > Br > I

Exam: 2003, 07

Soln: In going down a group the electronegativity values decreas and going from left to right in a period the electronegativity values increase. The halogens which are lie on the extreme right of the periodic table and most electronegative. Thus we see that the

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most electronegative element is F which occurs at the top right hand corner and the least electronegative element is cesium which occur at the bottom left hand corner of the periodic table.

Hence the electronegativity of halogens decrease in the order F > Cl > Br > I

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4.(c) Cu^{2+} complexes are coloured while those of Zn^{2+} are compounds the give below:

Cu^{2+} ($3d^9$, $n=1$) → blue cut ($3d^{10}$, $n=0$) → colourless ; Zn^{2+} ($3d^{10}$, $n=0$) → colourless.

Here n denotes the number of unpared electrons in the respective cations. It may be seen from the colours given that the cations Zn^{2+} having vacant or completely-filled d-orbitals (d^0 and d^{10} orbitals) aree colourles while those (e.g Ti^{3+}, V^{3+}, Cr^{3+} etc.) having partially filled d-orbitals (d^1, d^2, d^3 etc. orbitals) are coloured. In other words. the castions having all the electrons paired in d-orbitals (i.e. completely-filled d-orbitals) are colourless while those having some or all the electrons unpaired in d-orbitals are coloured.

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Mn^{2+} is more stable than Mn^{3+}; Fe^{3+} is more than Fe^{2+}

4.(f) The relative stability of various oxidation states of a given element can be explained on the basis of the stability of d^0, d^5 and d^{10} configurations. For example Ti^{4+} $(3d^0, 4s^0)$ is more stable than Ti^{3+} $(3d^1.4s^0)$, is more than Mn^{3+} $(3d^4.4s^0)$ since Mn^{2+} ion has $3d^5$ orbital. Ti^{4+} $(3d^0, 4s^0)$ is more stable than Ti^{3+} $(3d^1, 4s^0)$ is more than Fe^{3+} $(3d^4, 4s^0)$ since Fe^{2+} ion has $3d^5$ orbital. In going down a group, the higher oxidation states becom more stable. For example +2 and +3 oxidation states shown by Fe are stable oxidation states.

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(i) 4f-block elements: These elements are also called by other names like 1st inner-transition series elements, lanthanides and lanthanons.

(ii) 5f-block elements: Thes elements are also called by other names like 2nd inner-transition series elements, actinides and actinones.

What are lanthanides or rare-earth elements? Discuss their position in the periodic table.

Q. Position of lanthanides in the periodic table:-- All the fifteen lanthanides have atomic wooinghts between those periodic table. Barium has exactly the same outer electronic configuration as calcium and strontium and resembles them very closely. In a similar way hafnium is similar to zirconium (Z = 40). Therfore Ba must be placed below Sr (Group IIA) and Hf below Zr (Group IV B), Y (Z = 39, Group III B) thus leaving only one place between them. These elements also resemble Y39 because of the

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following additional similarities : (a) owing to lanthanide contraction (to be discussed subsequently), the ionic radius of Y^{3+} ion is almost similar that of Er^{3+} ($r_{Y^{3+}} = 0.93\text{Å}$, $r_{Er^{3+}} = 0.96\text{Å}$). (b) Y generally occurs in nature associated with the ores of heavier lanthanides and resembles Tb (III) and Dy (III) in its compounds : It, therefore, becomes necessary to accommodate all the fifteen lanthanides together at one place.

mention some uses of lanthanides.

Use of lanthanides and their compounds:-

(i) certain alloys of the lanthanide elements, known as mishmetals containing predominantly 300-35% of ce together with smaller quantities of other light

[P.T.O]

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lanthanides are used as reducing agents in metallurgical operations. mg-alloys containing about 3% misch metal and 1% Zr are used for making parts of jet engine.

(ii) La, Ce, Pr, Nd mixed with steel are used in cigarette lighters, toys, flame throwing tanks.

(iii) Nd_2O_3 and Pr_2O_3 are used as colouring agents for glass and in the production of standard filters. Lanthanum oxides are used for polishing glass. ceria (CeO_2) is used in gas mantles.

(iv) cerium phosphate is used as a catalyst in petroleum cracking.

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